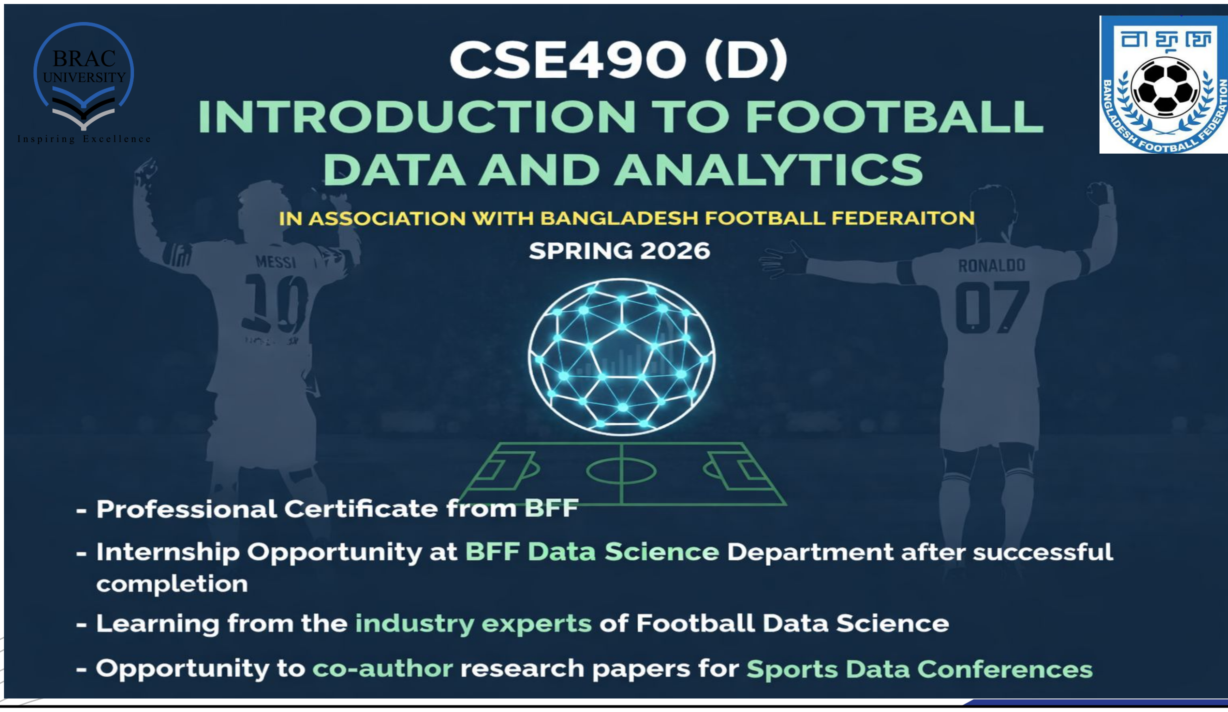

CSE490 (D)
Introduction to
Football Data & Analytics
Lecture-7

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Evaluating Actions

- We can use xG to measure the probability a goal is about to be scored.
- For other actions, things are more difficult.
- **Key idea: How does an action increase or decrease the probability of a team scoring**

Expected Threat (xT)

- Measure of the the affect of attacking actions
- **Position-based Expected Threat:**

  where we assign a value to every position on the field
- **Action-based Expected Threat:**

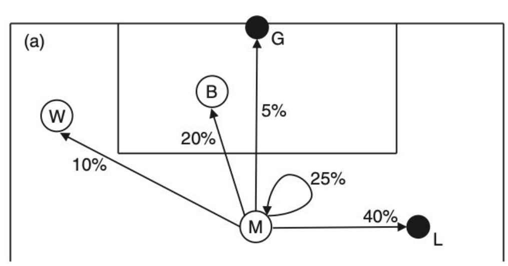
  where we assign a value to every action, based on its start and end position and various qualifiers describing that action

Markov Chain Framework

- The model divided the game in to a set of ‘states’
- Each state describes where the attacking team has the ball and the arrangement of the defending team.
- A Markov chain model assumes that the change between match states does not depend on what has happened earlier, only on the current state.
- This assumption doesn’t always hold in football, but it is a reasonable starting point.

Markov Chain Framework

- Fig. (a) states are represented as circles. Each of the percentage values indicates the probability that the next action is observed from when the ball is in midfield.
- Fig. (b) Probabilities of moving between states. Rows are the current state, columns are the next state, and the entries themselves are the probability of moving from one state to the next.
- The Goal and Lost states are special since they mark the end of the possession.
- These values are not measured from data, but are used to illustrate the approach.



(b)

From/to	Midfield	Box	Wing	Goal	Lost
Midfield	25%	20%	10%	5%	40%
Box	10%	25%	20%	15%	30%
Wing	10%	10%	25%	5%	50%
Goal	0%	0%	0%	100%	0%
Lost	0%	0%	0%	0%	100%

Markov Chain Framework

Who deserves credit for a goal?

Traditional football stats: Goal ⟶ Striker

Assist ⟶ Final passer

Problem: Earlier actions in the build-up are ignored.

Example sequence: Winger ⟶ Midfielder ⟶ Striker ⟶ Goal

Solution proposed by Sarah Rudd:

Assign credit based on **how much each action increases the probability of scoring.**

$$Credit = P_{after} - P_{before}$$

Markov Chain Framework

Way of solving –

1. Linear Algebra
2. Iterative Method
3. Monte Carlo Simulation

Follow the shared pdf and notebook.

Markov Chain Framework

Action	Probability Change	Credit
Winger → Midfielder	12% → 15%	+3
Midfielder → Striker	15% → 25%	+10
Striker scores	25% → 100%	+75

Still biased towards striker!

Markov Chain Framework

Players accumulate points across **all actions in a match**.

Example (Winger):

- 10 passes increasing probability by +3 → **30 points**
- 5 crosses increasing probability by +13 → **65 points**

Total = **95 points**

- Actions that reduce scoring probability → **negative points.** For example: pass from midfield to wing will result in **-3 points**.

Chance creation is rewarded, not just goals and assists.

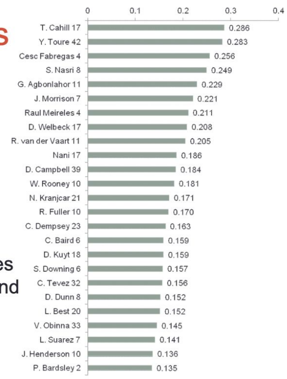
Markov Chain Framework

[redacted]

[redacted]

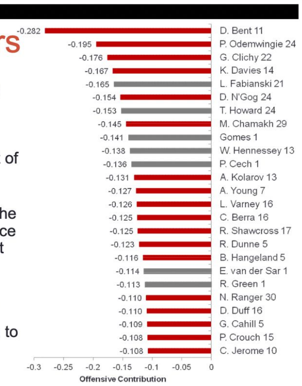
Top Performers

- Top performers are:
  - Tim Cahill
  - Yaya Toure
  - Cesc Fabregas
- Jordan Henderson in top 25
- Some surprises like James Morrison, Ricardo Fuller and Chris Baird



Worst Performers

- Lots of goal keepers, also strikers and defenders
- Poor Darren Bent
  - 1 goal was in sample set out of 17 he scored in the entire season
  - Had 19 opportunities where he had the ball with >10% chance of scoring (22% average) but only converted one
- Clichy/Young/Kolarov – poor crossers of the ball?
  - Can dig deeper into the data to identify which situations



Both were playing for Sunderland then. Bent was sold to Aston Villa for 24 million, Henderson went to Liverpool for less than 20 million!

Research Proposal (15% marks)

1. What are you trying to solve?
2. The dataset you are going to use.
3. The model(s) you are planning to use.

Deliverables:

- 3 minutes presentation.
- 2/3 reference papers.
- Dataset link.

Date of presentation: **16.03.2026**